

Challenges Faced

- **Dataset Availability:** A suitable stock market dataset was not initially available, so a sample dataset containing daily stock prices was prepared for the practical.
- **Spark Output Logs:** Spark generated a large number of system logs during execution, which made the actual analysis output difficult to identify. This was handled by setting the Spark log level to `ERROR`.
- **Data Type Handling:** The stock dataset contained date and numerical price fields, so the schema was checked carefully to ensure that the data was loaded with appropriate data types.
- **Window Functions:** Calculating daily returns and moving averages required the stock records to be properly ordered by date and grouped by stock symbol.
- **Handling Initial Values:** The first record of each stock does not have a previous day's price, so its daily return resulted in a `NULL` value.
- **Visualization in WSL:** The Matplotlib chart window did not remain open properly in the WSL environment. The chart was therefore saved directly as a PNG file for viewing and reporting.
- **Large Data Processing Concepts:** Understanding grouping, aggregation, window functions, returns, and volatility helped demonstrate how Spark can be applied to financial data analysis.